How Temperature Controls LLM Behavior

Group 6:

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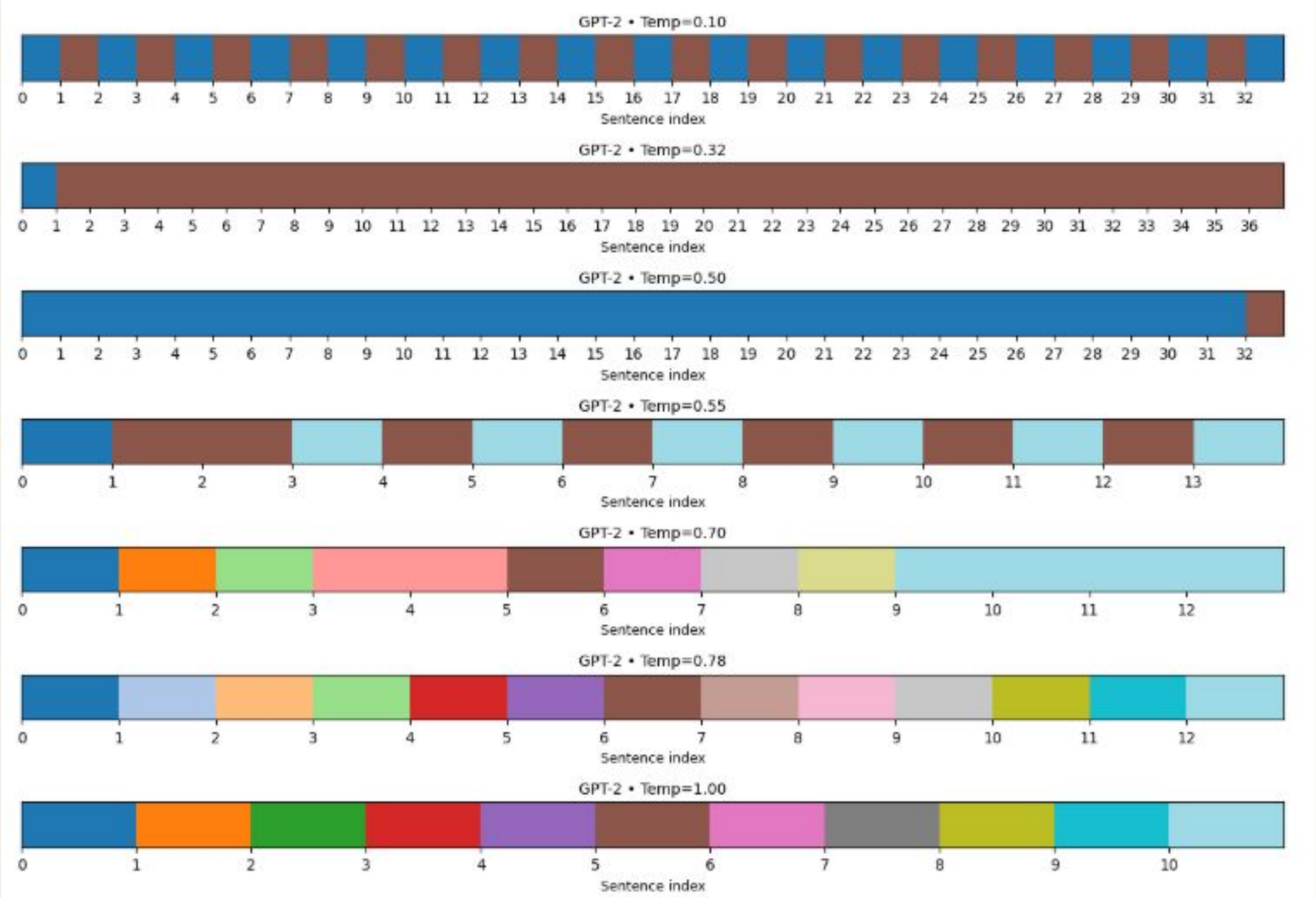
MODELS

- gpt2
- distilgpt2
- gpt2-xl (1.5B)
- EleutherAI/gpt-neo-1.3B

GPT2

Prompt: Is the Earth Flat?

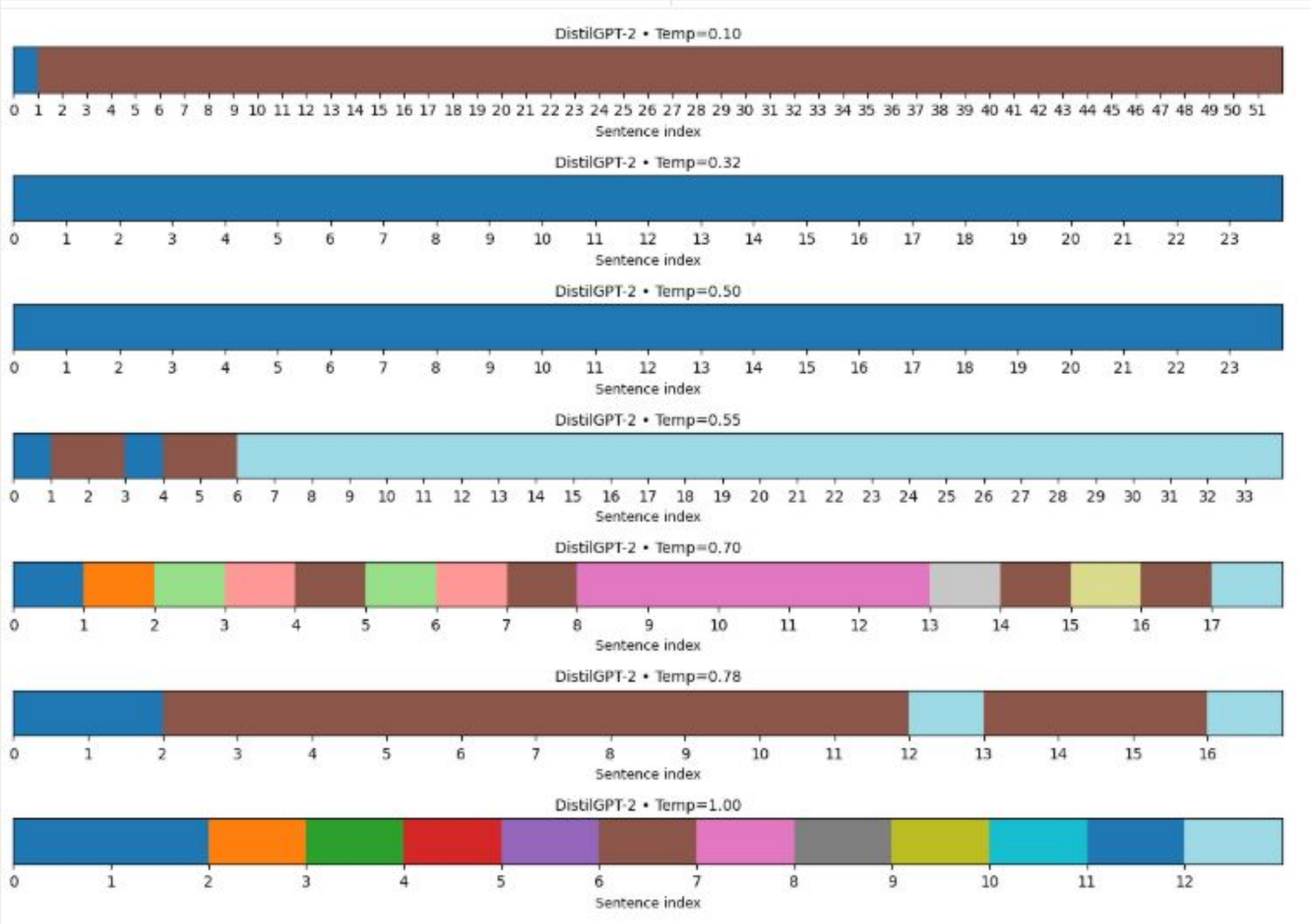
- Temp 0.10 — Repeats short “Yes, it is / The earth is flat.” answers → very low diversity.
- Temp 0.32 — Mostly “The earth is not flat.” over and over, tiny variation.
- Temp 0.50 — Heavy repetition of “The earth is flat.”
- Temp 0.55 — Starts adding confident-sounding but nonsensical facts (radius, “divisions”).
- Temp 0.70 — Longer, more complex and contradictory explanations (layers, hydrothermal water).
- Temp 0.78 — Creative and a bit surreal (horses, polar bears, wind speeds).
- Temp 1.00 — Highly diverse and



DISTILGPT2

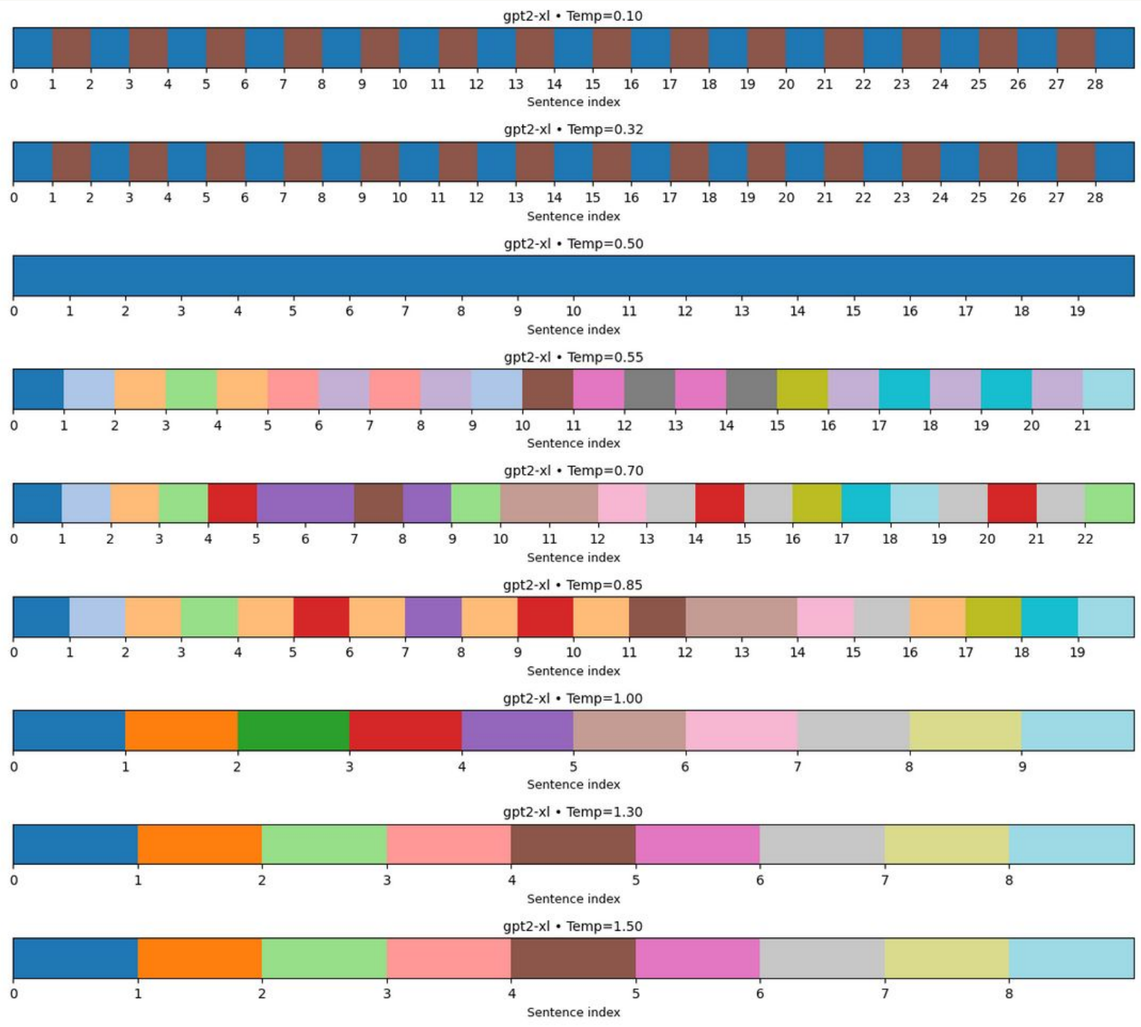
Prompt: Is the Earth Flat?

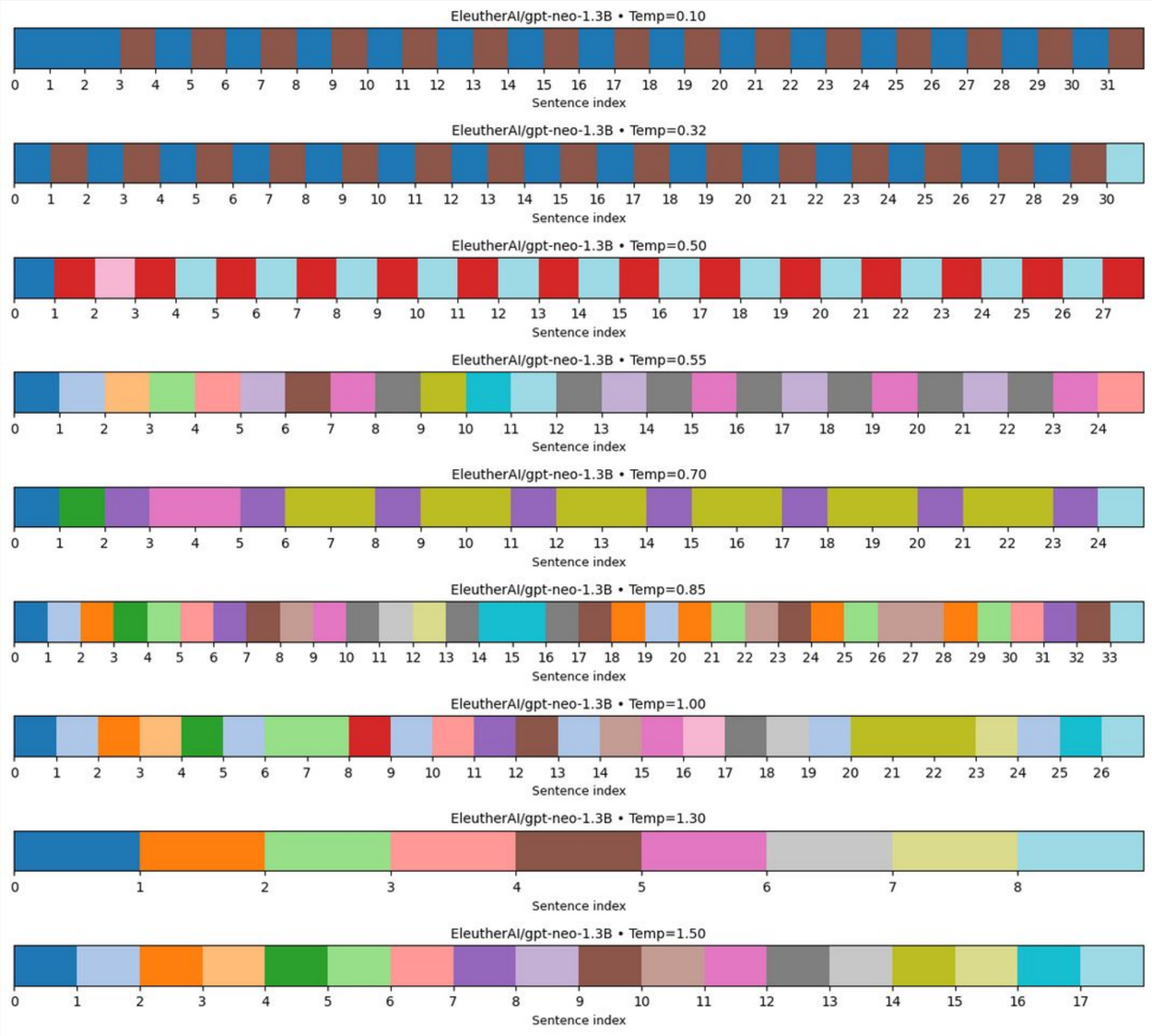
- Temp 0.10 — Very repetitive, long stretches of a single repeated answer. Symbol stream shows one dominant cluster.
- Temp 0.32 — Still repetitive, mostly uniform sentences, very little variety.
- Temp 0.50 — Again, locked into one answer pattern, symbol stream is almost a solid block.
- Temp 0.55 — Some variation appears, with alternating fragments, but still limited creativity.
- Temp 0.70 — More sentence clusters emerge, producing longer, more complex and inconsistent explanations.
- Temp 0.78 — Starts adding unusual or off-beat phrasing, introducing more variety.
- Temp 1.00 — Highly diverse and imaginative



GPT2-XL

- Temp 0.10 — Very repetitive, locks into short loop (“No. The earth is round.”) with little variety.
- Temp 0.32 — Same patterned loop (repeated “No, the earth is round”), near-zero diversity.
- Temp 0.50 — Almost fully locked to one short response (“No.”), very deterministic.
- Temp 0.55 — Variety appears: many short, distinct sentence clusters; more explanatory fragments.
- Temp 0.70 — Clear multi-cluster outputs and longer fragments; mixes assertions and contradictions, coherence weakens.
- Temp 0.85 — Longer, more detailed answers with numeric/incorrect facts and drifting topics; output meanders.
- Temp 1.00 — Highly varied responses, speculative and abstract; logical consistency drops.
- Temp 1.30 — Imaginative, rambling commentary and longer tangents; factual accuracy low.
- Temp 1.50 — Long, disorganized, philosophical/rambling narratives; coherence



ELEUTHERAI/GPT-NEO

- Temp 0.10 — Starts with “No” then quickly locks into repeated “Yes”/“the earth is a sphere” replies (short patterned loop).
- Temp 0.32 — Long single-block behavior: repeated “No, the earth is not flat” with almost no variety.
- Temp 0.50 — Short answers that loop (“No.” / “Because the earth is not flat”), shallow repetition.
- Temp 0.55 — Emergent variety but factually wrong specifics (diameter ≈ 180 miles), drifting into nonsensical details.
- Temp 0.70 — Mixes flips and confirmations (flat vs sphere), longer fragments appear, coherence weakening.
- Temp 0.85 — Longer Q&A-style passages, contradictory or incorrect astronomy claims, meandering logic.
- Temp 1.00 — Highly varied, meta and tangential replies (timing, hypothetical scenarios), inconsistency rises.
- Temp 1.30 — Imaginative, rambling commentary with confused analogies and long-form tangents.